

In how many different ways can 2 red, 2 green and 2 blue balls be arranged in a row?

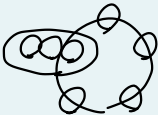
$6C_2 \times 4C_2 = 90$

Answer:

Four boys and three girls are to be arranged at a round table. In how many ways can this be done, if the girls want to sit next to each other, and two arrangements are equivalent if one of them can be rotated into the other.

$\frac{5P5}{5} \times 3P3$

Answer:



In how many ways can we read the word COMPUTER from the table below? (We can step only to neighbouring letters).

C O M P U
O M P U T
M P U T E
P U T E R

$\rightarrow \rightarrow \rightarrow \rightarrow \downarrow \downarrow \downarrow$
 $7C3$

Answer:

Five children, including a pair of siblings, are to be arranged in a row. In how many different ways can this be done, if the younger sibling wants to stand right after his brother?

$000 @ 9 \quad 4P4$

Answer:

Six friends agree to meet this evening. Everybody brings a bottle of wine, however, all of them buys the wine in the same small shop, where only 4 kinds of wine can be bought. How many different wine collections can they have when they meet?

$4+6-1 C 6 = 9C6$

Answer:

In a box there are 10 cards, with the digits 0, 1, ..., 9, respectively. We draw 4 cards, one by one, and write down their numbers. How many different sequences can we get?

$\underline{10} \quad \underline{9} \quad \underline{8} \quad \underline{7} \quad 10C1 \times 9C1 \times 8C1 \times 7C1$

Answer:

In a company there are 4 girls and 5 boys. In how many different ways can we choose 4 people to have exactly 2 girls among the chosen people?

$4C2 \times 5C2$

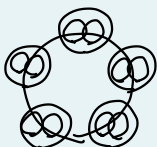
Answer:



Five pairs of siblings are to be arranged at a round table. In how many ways can this be done, if the siblings want to sit next to each other, and two arrangements are equivalent if one of them can be rotated into the other.

$$\frac{{}^5P_5}{5} \times 2^5$$

Answer: 768



How many 4-digit numbers do exist, where all the digits are odd?

$$\underline{5} \underline{5} \underline{5} \underline{5} = 5^4 = 625$$

Answer: 5 625

$$\begin{matrix} 1 & 3 & 5 & 7 & 9 \\ 0 & 2 & 4 & 6 & 8 \end{matrix} \leftarrow 4 \text{ digits} : 5C4$$

There are 10 balls in an urn, numbered from 1 to 10. We draw 5 balls. How many cases are where exactly on 3 drawn balls stands a number less than 5?

$$\begin{matrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 & 9 & 10 \end{matrix} \leftarrow 3 \text{ balls} : 4C3 \quad 4C3 \times 6C2$$

Answer: 60

How many different character sequences with length 3 can be formed using the letters A, B, C, D, E, F?

$$\underline{6} \underline{5} \underline{4} \quad 6C1 \times 5C1 \times 4C1$$

Answer: 120

There are 20 red, 20 blue and 20 green balls in an urn. We take out 7 balls. How many distributions can the colours of drawn balls have?

$$3+7-1 \ C \ 7 = 9C7$$

Answer: 36

We have 6 cards with the digits 1, 1, 2, 2, 2, 3, respectively. How many 6-digit numbers can be formed using this cards?

$$\begin{matrix} 1 & 1 \\ 2 & 2 & 2 \\ 3 \end{matrix}$$

Answer: 60

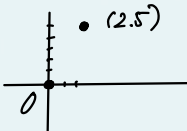
$$6C1 \times 5C2$$

Seven children, including a pair of siblings, are to be arranged in a row. In how many different ways can this be done, if the younger sibling wants to stand right after his brother?

$$\circ \circ \circ \circ \circ \circ \circ \quad 6P6$$

Answer: 720

Find the number of possible paths from the origin to the point (2, 5) if we can walk only on points with integer coordinates and we can step only upwards and right.



$$\rightarrow \rightarrow \uparrow \uparrow \uparrow \uparrow \uparrow$$

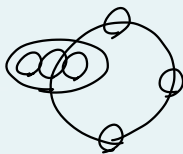
$$7C2$$

Answer: 21

Three boys and three girls are to be arranged at a round table. In how many ways can this be done, if the girls want to sit next to each other, and two arrangements are equivalent if one of them can be rotated into the other.

$$\frac{4P_4}{4} \times 3P_3$$

Answer: 36



A biscuit factory hides a plastic figure in every box of biscuits. There are 4 kinds of figures and somebody buys 8 boxes of biscuits. How many possible distributions of figures he can find in the boxes?

$$4+8-1 \ C \ 8 = 11C_8$$

Answer: 165

There are 10 balls in an urn, numbered from 1 to 10. We draw 4 balls. How many cases are where exactly on 2 drawn balls are prime numbers?

Answer: 90

$$\begin{cases} 2 & 3 & 5 & 7 \\ 1 & 4 & 6 & 8 & 9 & 10 \end{cases} \leftarrow \begin{matrix} 2 \text{ num} : 4C_2 \\ 2 \text{ num} : 6C_2 \end{matrix} \quad 4C_2 \times 6C_2$$

Six children, including a pair of siblings, are to be arranged in a row. In how many different ways can this be done, if the younger sibling wants to stand right after his brother?

$$0 \ 0 \ 0 \ 0 \ @ \ 5P_5$$

Answer: 120

Rolling a dice three times, how many cases are, where all results are even numbers?

$$\begin{cases} 2 & 4 & 6 \\ 1 & 3 & 5 \end{cases} \leftarrow \underline{3} \ \underline{3} \ \underline{3} = 3^3$$

Answer: 27

In how many different ways can 2 red and 4 blue balls be arranged in a row?

$$6C_2$$

Answer: 15

A salesman plans to visit 5 cities during the next 5 days (one city per day). In how many ways can he organize his next three days?

$$\downarrow \quad A \ B \ C \ D \ E \quad \underline{5} \ \underline{4} \ \underline{3} \quad 5C_1 \times 4C_1 \times 3C_1$$

Answer: 60

In how many ways can we read the word COMPUTER from the table below? (We can step only to neighbouring letters).

$$\begin{matrix} C & O & M & P & U & T \\ O & M & P & U & T & E \\ M & P & U & T & E & R \end{matrix} \quad \begin{matrix} \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \downarrow \downarrow \\ 7C_2 \end{matrix}$$

Answer: 21

Eight children, including a pair of siblings, are to be arranged in a row. In how many different ways can this be done, if the younger sibling wants to stand right after his brother?

$00000000 @ \rightarrow$

Answer: 5040

Peter has forgotten his 4-digit PIN. He remember, that it has four different digits and the second digit is 3. How many different possibilities are for his PIN?

012456789
 $- \underline{3} - -$
 $9C_1 \times 8C_1 \times 7C_1$

Answer: 504

How many 4-digit numbers do exist where all the digits are even?

$\begin{Bmatrix} 0 & 2 & 4 & 6 & 8 \\ 1 & 3 & 5 & 7 & 9 \end{Bmatrix}$

Answer: 500

$4 \underline{5} \underline{5} \underline{5}$

In a box there are 7 cards, with the characters A, A, B, B, C, C, C, respectively. We draw the cards one by one and write down the characters in order. How many different character sequences can we get?

$\begin{Bmatrix} AA \\ BB \\ CCC \end{Bmatrix} \quad 7C_2 \times 5C_2$

Answer: 210

We have 10 boxes, in every box a chewing gum, a chocolate, or a lollipop. How many different collections can we have, when we open the boxes?

$3+10-1 \ C_{10} = 12C_{10}$

Answer: 66

On the real axis, starting from the origin we can always step 1 right or left. In how many ways can we arrive in the point +2, if the total number of steps is 10?

$\begin{matrix} \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \rightarrow \\ \leftarrow \leftarrow \leftarrow \leftarrow \end{matrix} \quad 10C_4$

Answer: 210

There are 10 balls in an urn, numbered from 1 to 10. We draw 4 balls. How many cases are where exactly on 1 drawn ball is a prime number?

$\begin{Bmatrix} 2 & 3 & 5 & 7 \\ 1 & 4 & 6 & 8 & 9 & 10 \end{Bmatrix} \begin{matrix} \leftarrow 4C_1 \\ \leftarrow 6C_3 \end{matrix} \quad 4C_1 \times 6C_3$

Answer: 40 80

$4C_1 \times 5C_3$

Five boys and three girls are to be arranged at a round table. In how many ways can this be done, if the girls want to sit next to each other, and two arrangements are equivalent if one of them can be rotated into the other.

$\frac{6P_6}{6} \times 3P_3$

Answer: 720

